

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL			
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLTM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 APR 2024	First Release	Yeap Leong Wei

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Content

Content.....	2
Scope	3
Tool List	3
Origin X-Axis Encoder Head Removal Procedure	5
New X-Axis Encoder Head Cable Replacement Procedure	13
Tool Removal and Cleaning	18
Reference for Alignment Procedure	18

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Scope

This procedure provides guidelines for the removal and replacing the X-Axis Quantic Encoder Read Head.

Tool List

No	Part No	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Metric T-Handle Wrench (M5 size)	1	PC	
3	-	Camera Tool	1	PC	
4	-	Temper Proof Screw Driver Set	1	Set	
5	-	Cutter	1	PC	

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

6	4XASPP-A440	Renishaw Quantic Encoder Alignment Kit	1	Set	
---	-------------	--	---	-----	--

Table 1: Tool List

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Origin X-Axis Encoder Head Removal Procedure

1. **Board Unloading:** Prior to commencing the removal procedure, ensure that any boards within the machine are safely unloaded. Follow the manufacturer's guidelines or equipment manual for proper board removal procedures.
2. **Shutdown Sequence:** Close V810 GUI and initiate the shutdown sequence for the Super Micro PC by following the manufacturer's recommended procedure. Ensure all data is saved and any necessary backups are performed before proceeding.
3. **Deactivation of Equipment:** At the operator console panel, press the “X-Ray Off” button to deactivate any X-ray equipment. Turn the key to the “Off” position to ensure complete deactivation of the system and associated components.
4. **Power Down Procedure:** Safely power down the SSCA main power source by following the manufacturer's recommended shutdown procedure. Ensure all power indicators are off before proceeding with the removal process.
5. **Access Door Opening:** Open all front access doors carefully to provide sufficient access to the internal components of the machine. Ensure doors are fully opened and secured to prevent accidental closure during the removal process.
6. **Left Outer Barrier Removal:** Remove the left outer barrier by unscrewing the door screws using the appropriate tools. Store the screws in a secure location for reassembly after the removal process is complete.

a) For S2EX & XLT



Figure 1: Left Outer Barrier (S2EX & XLT)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

b) For S3

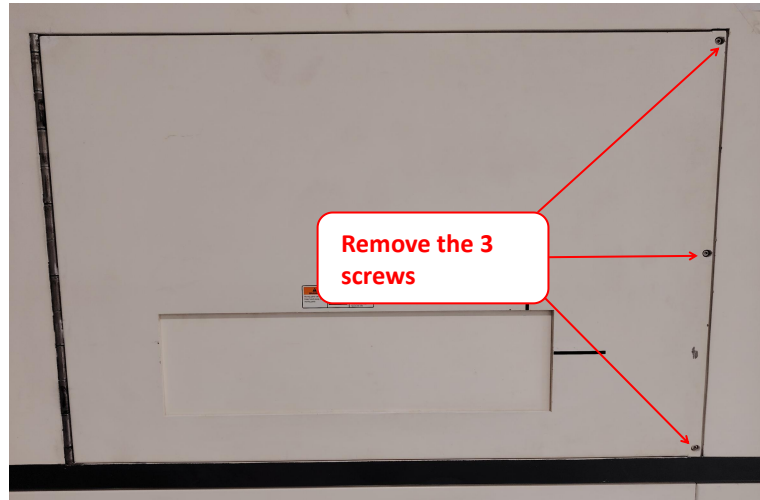


Figure 2: Left Outer Barrier (S3)

7. **Rear Access Door Opening:** Open the rear access door carefully, ensuring there are no cables or obstructions that may hinder the removal process. Take caution to prevent strain on any cables or connectors located near the access door.
8. **Cable Tie Removal:** Back to machine front side, remove any cable ties from the cable tray that may be securing the encoder cables. Use cable tie removal tools if necessary to avoid damage to the cables or surrounding components.



Figure 3: X-Axis Read Head Encoder cable routing

9. **X-Axis Motor Cable Disconnection:** Disconnect the X-Axis motor cable connector from its corresponding port. Remove any screws securing the cable to the cable tray and set them aside for reassembly.

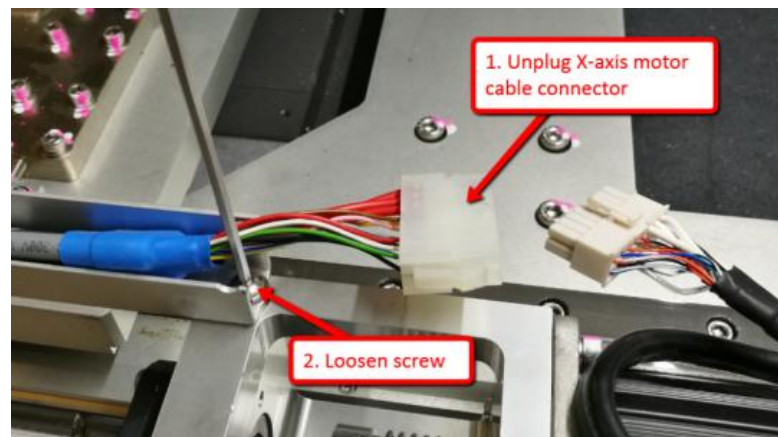


Figure 4: X-Axis Rotary Motor (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

10. **X-Axis E-Chain Guard Plate and Bracket Removal:** Unfasten the screws securing the X-Axis E-chain guard plate and motor cable bracket. Carefully remove these components and store them in a safe location for re-installation.

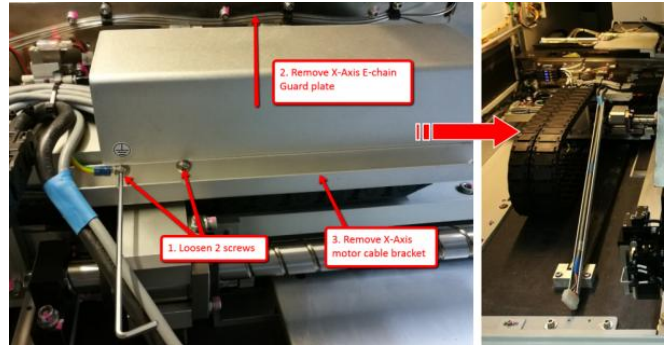


Figure 5:X-Axis Cable Tray and E-Chain Guard Plate (Sample Image)

11. **Cable Clamp Removal:** Unfasten the screws securing the cable clamp and remove it from the assembly. Take care not to damage the cables or surrounding components during removal.



Figure 6:X-Axis Read Head Encoder (Sample Image)

12. **X-Axis Read Head Encoder Bracket Removal:** Using a camera tool for precision, unfasten the screws securing the X-Axis read head encoder bracket. Carefully remove the encoder, ensuring gentle handling to prevent any damage.

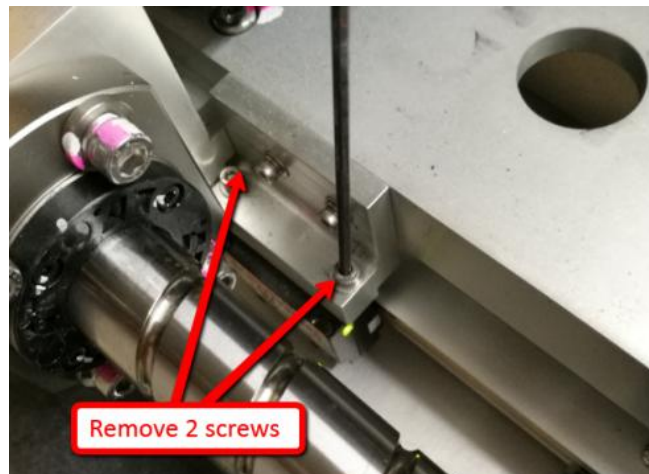


Figure 7:X-Axis Read Head Encoder Bracket (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

13. **X-Axis E-Chain Slot Cover Opening:** Use a flat-head screwdriver to carefully open the first X-Axis E-chain slot cover. Apply gentle pressure to avoid damaging the cover or surrounding components.

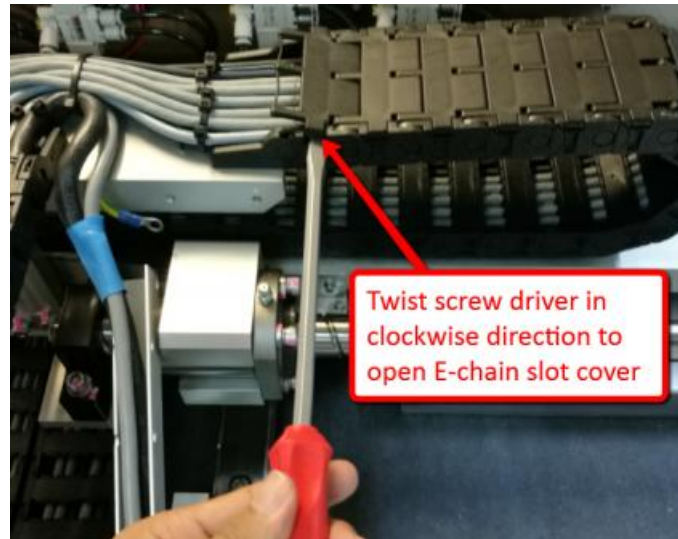


Figure 8: X-Axis E-Chain

14. **Cable Tie Removal:** Remove any necessary cable ties from the cable tray, ensuring that cables are free and unobstructed for removal. Take care to avoid damaging the cables or surrounding components.

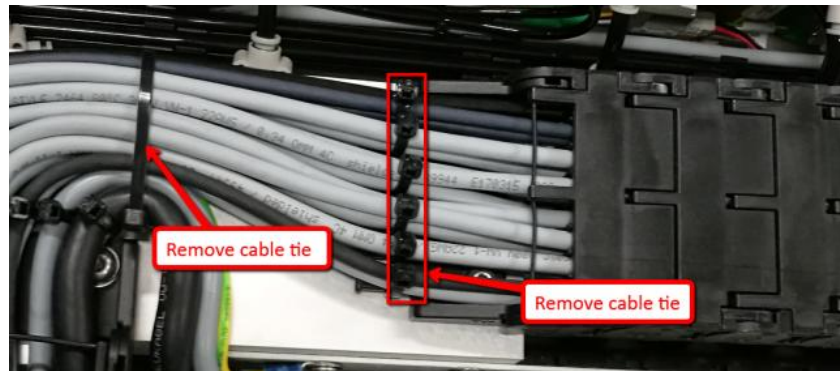


Figure 9: X-Axis E-Chain (Partial cover removed)

15. **X-Axis E-Chain Top Joint Opening:** Use a flat-head screwdriver to carefully open the X-Axis E-chain top joint. Ensure smooth movement and avoid any strain on the cables or connectors.

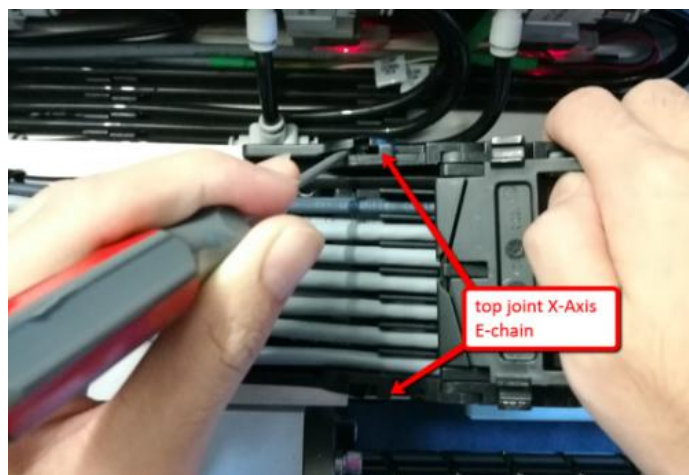


Figure 10: X-Axis E-Chain (Top Joint)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

16. **Linear Encoder Cable Removal:** Loosen the screws securing the linear encoder cable and carefully remove it from the assembly. Pay attention to the cable routing to ensure smooth removal without causing any damage.

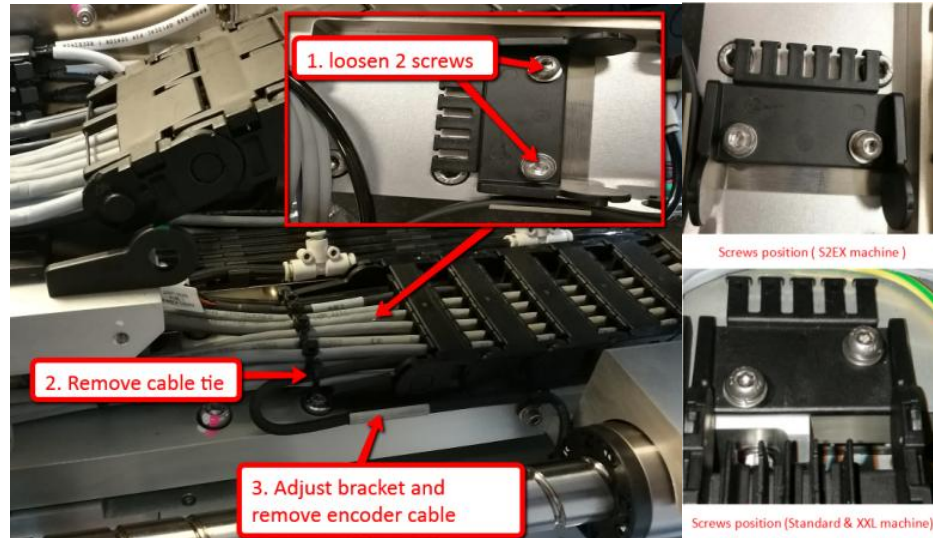


Figure 11: X-Axis E-Chain

17. **X-Axis E-Chain Bottom Joint Opening:** Use a flat screwdriver to gently open the X-Axis E-chain bottom joint. Take care not to exert excessive force to prevent damage to the cables or connectors.

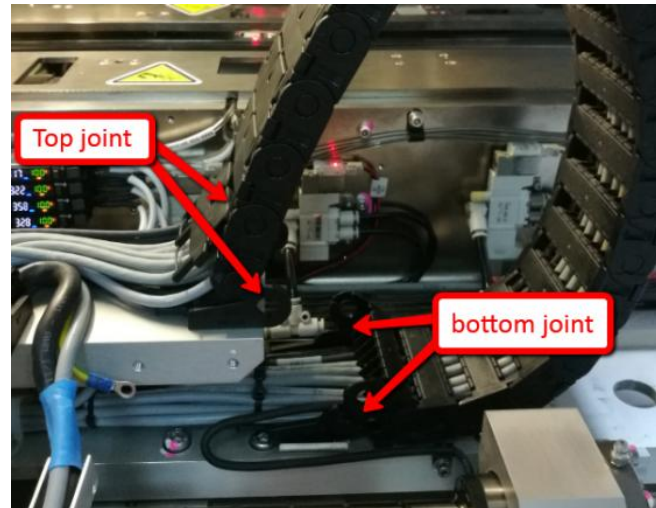


Figure 12: X-Axis E-Chain (Top & Bottom Joint)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

18. **Remaining X-Axis E-Chain Slot Cover Opening:** Open the remaining X-Axis E-chain slot covers systematically, applying cable ties after opening every 2-3 slot covers to maintain organization and prevent tangling.

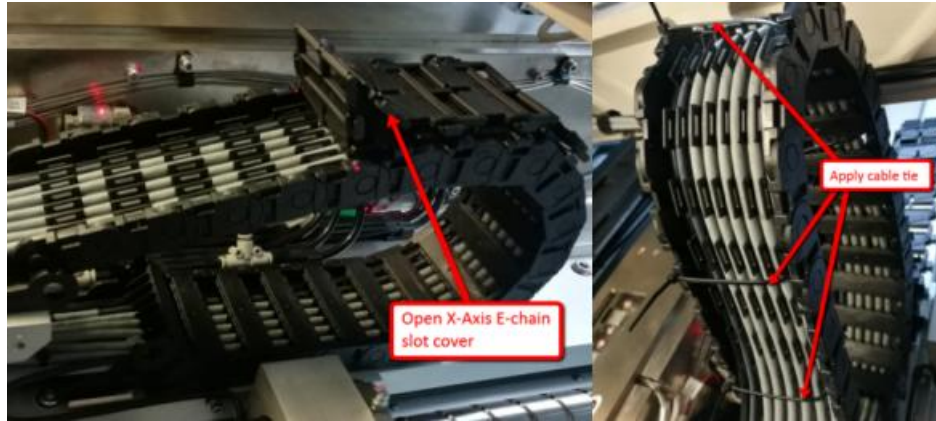


Figure 13: X-Axis E-Chain

19. **X-Axis Read Head Encoder Cable Removal:** Gently remove the X-Axis read head encoder cable from the X-Axis E-chain, ensuring smooth movement and avoiding any damage to the cables or connectors.
20. **Additional Cable Tie Removal and Y-Axis E-Chain Slot Cover Opening:** Remove any necessary cable ties, using a flat-head screwdriver to open end joints and slot covers on the Y-Axis wide E-chain (right-hand side). Push the stage to the rear side if needed for easier access.



Figure 14: Y-Axis E-Chain

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

21. **X-Axis Linear Encoder Cable Removal:** Gently remove the X-Axis linear encoder cable from the X-Axis E-chain, ensuring smooth movement and avoiding any damage to the cables or connectors.

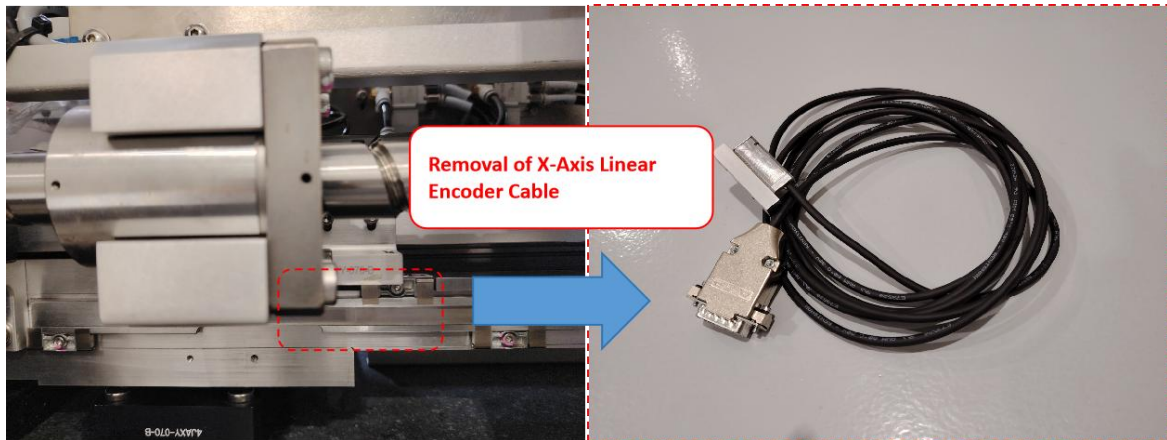


Figure 15: Removal of X-Axis Encoder Head Cable

22. **Dismantle the Read Head Encoder from the Bracket:** Unfasten the 2 screws securing the read head encoder to the bracket. Carefully place the bracket in visible place.

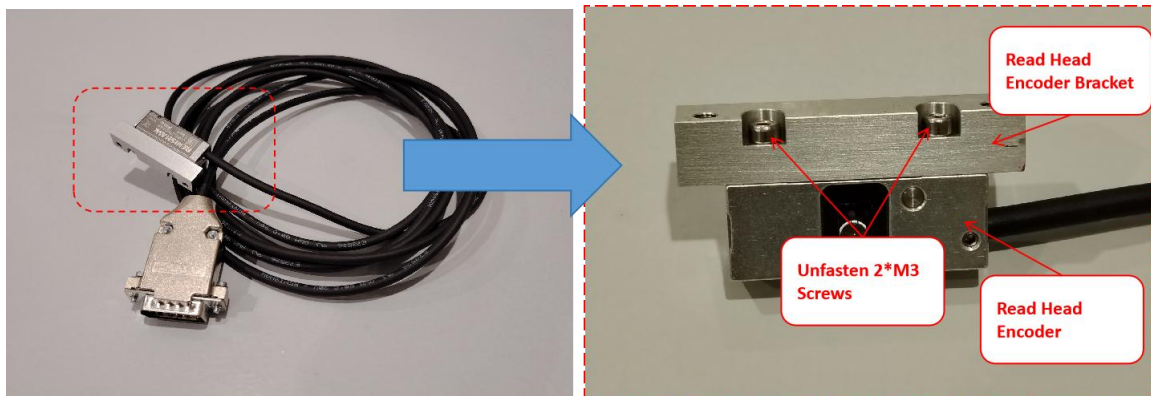


Figure 16: Dismantle X-Axis Encoder Head from Bracket

23. **Remaining Y-Axis E-Chain Slot Cover Opening and Cable Tie Application:** Open the remaining Y-Axis wide E-chain slot covers systematically, applying cable ties only to non-related cables to maintain organization and prevent tangling.
24. **X-Axis Encoder Cable Removal from Y-Axis E-Chain:** Gently remove the X-Axis encoder cable from the Y-Axis E-chain, ensuring that it is free from any entanglement or obstruction.

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

25. **Read Head Encoder Cable Disconnection:** Disconnect the read head encoder cable from the extension cable connected to the SSCA. Take care to avoid any strain on the connectors or cables.

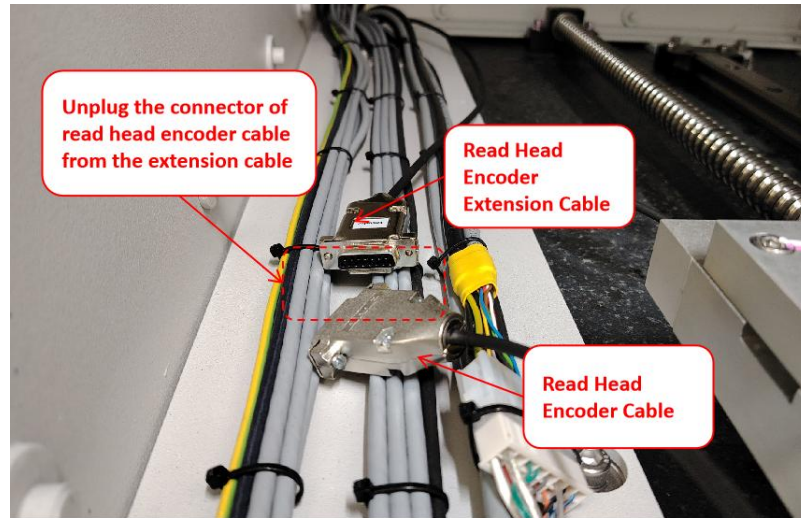


Figure 17: Disconnection for the Extension of X-Axis Encoder Cable

26. **Final X-Axis Encoder Removal:** Gently remove the X-Axis encoder from the machine, ensuring that it is handled carefully to prevent any damage during removal and transport.

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

New X-Axis Encoder Head Cable Replacement Procedure

27. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

Part number	Part Description	QTY	UOM	Illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	

Table 2: Encoder Head, Optical, 0.5UM

28. **Encoder Assembly:** Assemble the X-Axis Encoder to the origin mounting bracket.

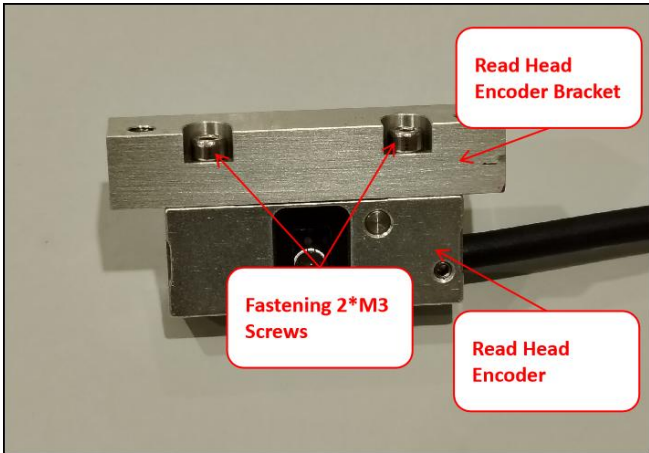


Figure 18:Securing X-Axis Read Head Encoder to Bracket

29. **Encoder Bracket Assembly:** Securely tighten the two screws to ensure proper attachment.

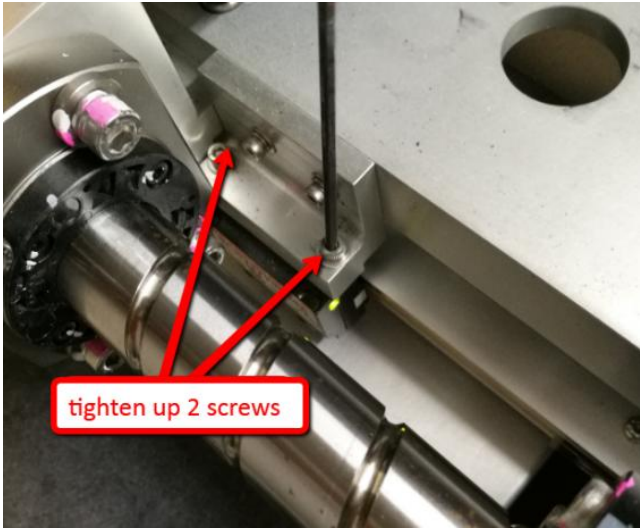


Figure 19:Attaching X-Axis Read Head Encoder Bracket

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

30. **Cable Clamp Application:** Apply the cable clamp to secure the encoder cables in place and tighten the two screws firmly.



Figure 20: X-Axis Ball Screw Bearing (Sample Image)

31. **Read Head Encoder Cable Installation:** Securely clamp back the X-Axis read head encoder cable and tighten the two screws to ensure stable connection.

Note: Ensure the cable holder is full press inside before securing the screws.

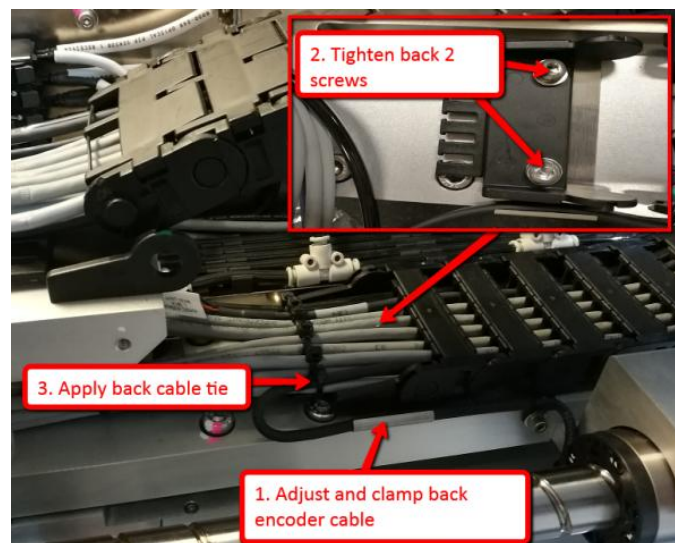


Figure 21: X-Axis E-Chain (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

32. **Cable Routing and E-Chain Closure:** Gently route the X-Axis encoder cable back to its original position, removing any necessary cable ties for flexibility. Close the E-chain slot cover to each partition securely.

Note: If machine using the Tribo Tape mode, perform cable management nicely to avoid cable twisted.



Figure 22:X-Axis E-Chain (Sample Image)

33. **E-Chain End Joint Closure and Cable Management:** Close the E-chain end joints on both sides and apply necessary cable ties to maintain proper cable organization and routing.

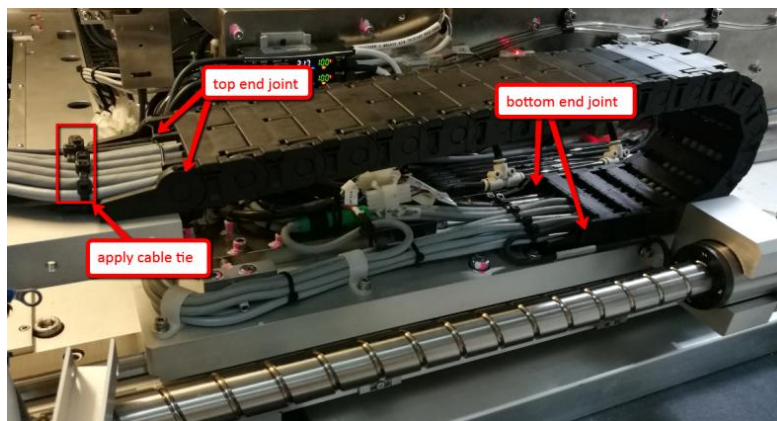


Figure 23:X-Axis E-Chain End Joint (Top & Bottom)

34. **Stage Movement and Cable Inspection:** Push the X-Axis stage to the left and right several times to ensure smooth movement. Verify that cables are well-routed within the E-chain. If any issues arise, open the E-chain and re-route the cables as needed.

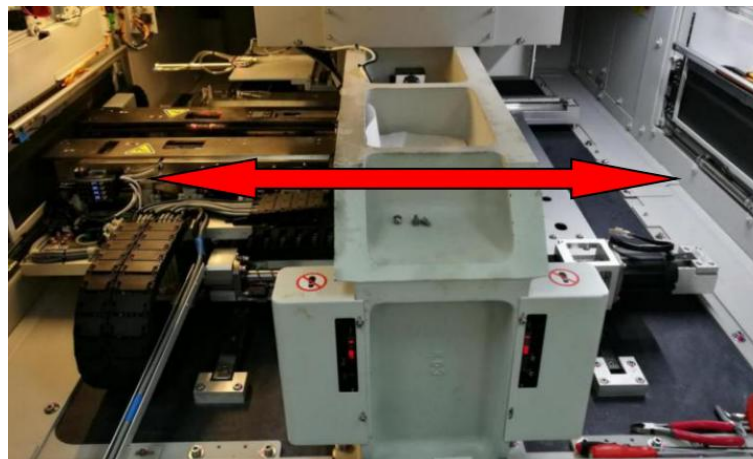


Figure 24:X-Axis Stage (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

35. **Bracket and Guard Plate Assembly:** Assemble the X-Axis motor cable bracket and guard plate, securely tightening the two screws. Note that the grounding wire is screwed together with the guard plate for proper grounding.

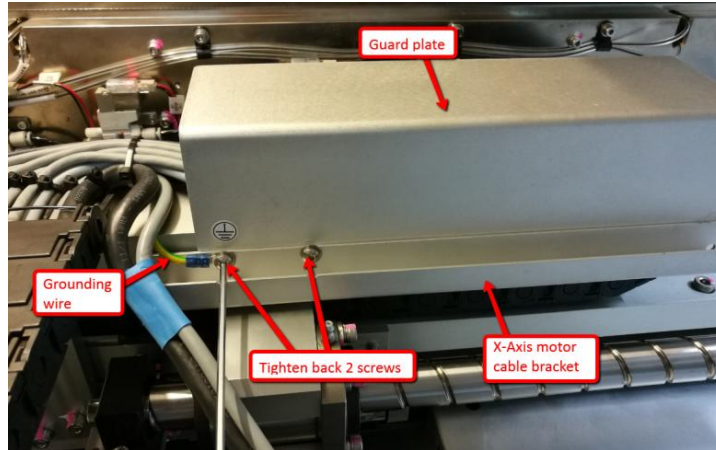


Figure 25: X-Axis Cable Tray and Guard Plate

36. **Motor Cable Connector Re-connection:** Tighten the screw and plug back the X-Axis motor cable connector to restore electrical connection.

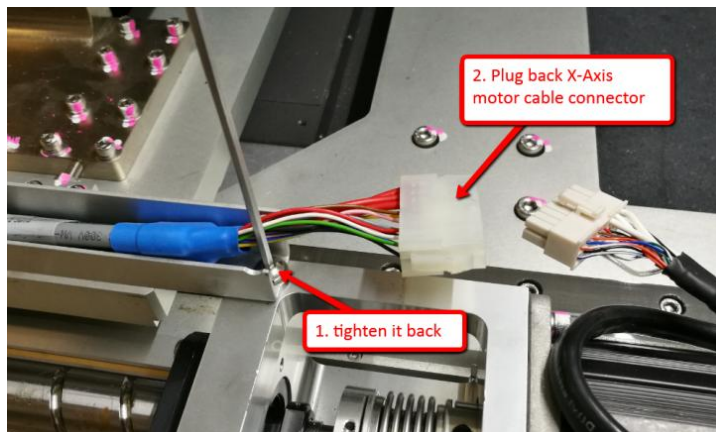


Figure 26: X-Axis Rotary Motor Housing

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

37. **Encoder Cable Routing to Y-Axis E-Chain:** Gently route the X-Axis encoder cable to the Y-Axis E-chain. Close the E-chain slot covers to each partition according to the provided diagram. Ensure all cables are not twisted to prevent interference.

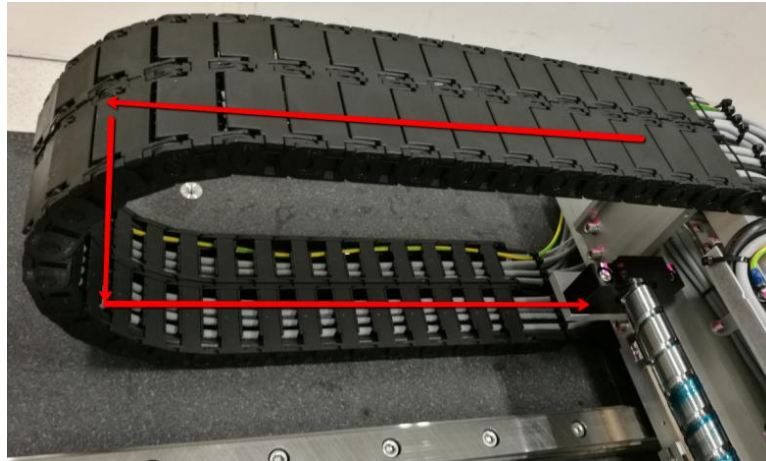


Figure 27: Y-Axis E-Chain

38. **Y-Axis E-Chain Closure and Cable Management:** Close the end joints on both sides of the Y-Axis E-chain and apply necessary cable ties to maintain proper cable organization and routing.

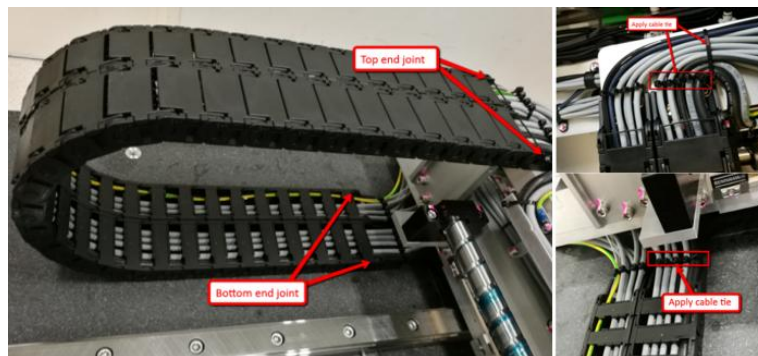


Figure 28: Y-Axis E-Chain End Joints (Top & Bottom)

39. **Stage Movement and Cable Inspection (Y-Axis):** Move the Y-Axis stage to the front and rear several times to ensure smooth movement. Verify that cables are well-routed within the E-chain. If any issues arise, open the E-chain and re-route the cables as needed.

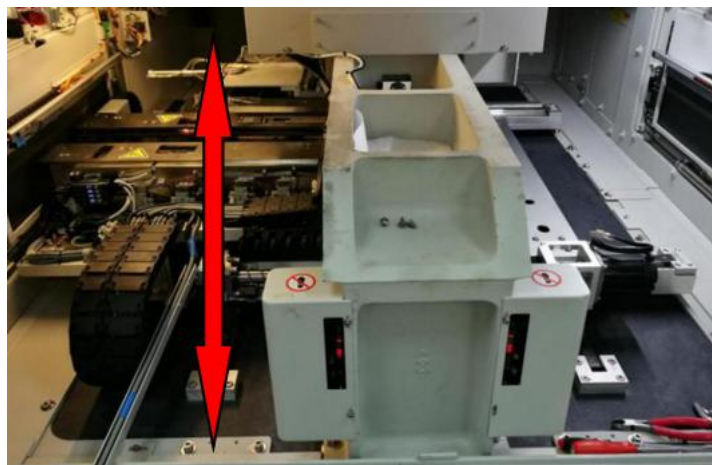


Figure 29: Y-Axis Stage

	Work Instruction	Doc. Revision	00
	AXI V810 X-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

40. **Read Head Encoder Connection:** Reconnect the read head encoder with the extension cable to the SSCA to restore full functionality.

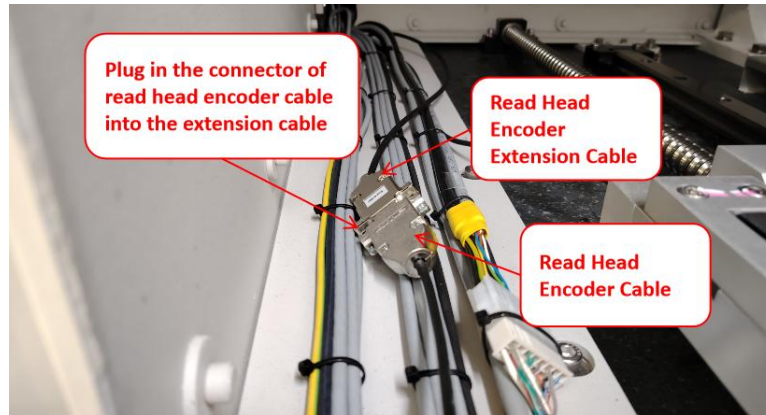


Figure 30: X-Axis Encoder Cable Connection

Tool Removal and Cleaning

41. Before concluding the replacement procedure, carefully inspect the machine interior to ensure that all tools, equipment, and debris have been removed. Retrieve any tools or equipment left inside the machine and return them to their designated storage area to prevent any interference with machine operation.
42. Dispose of any removed cable ties, packaging materials, or other debris in the designated waste disposal containers according to company guidelines or environmental regulations.
43. Once the interior surfaces have been cleaned and cleared of debris, visually inspect the machine to ensure that it is in a tidy and organized state, ready for operation.
44. Perform a final check to verify that all components have been properly installed and secured according to manufacturer specifications. Confirm that all cables are properly routed and secured within their respective E-chains to prevent any interference with machine movement.
45. Once the machine has been cleaned and inspected, power on the system according to standard operating procedures but without launch the V810 GUI.

Reference for Alignment Procedure

46. **Encoder Read Head Alignment:** Refer to “AXI V810 Quantic Encoder Read Head Alignment Procedure” to verify the encoder status and perform necessary alignment if needed.

Note: Crucial to ensure the signal strength for encoder read head along the x-axis stroke.